

Alexandru Ardelean

Embedded SW/Linux Kernel Engineer

Contact details

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Location: Cluj-Napoca

About Me

Embedded software developer, primarily skilled in C, shell, some Python, some automation, and some various other topics. Contributor to various open-source projects (Linux kernel, OpenWrt, and other).

Github profile: <https://github.com/commodo/>

Linux kernel contributions: roughly 430+ authored (with various email addresses)

- <https://github.com/torvalds/linux/commits?author=commodo> or
- `git log --oneline --author="Alexandru Ardelean" | wc -l`

More recently, I started doing Computer Vision with AI model training (YOLO, HRnet) and getting them to work on embedded devices.

Skill sets

C	15+ years	Python	6+ years
Embedded Development	15+ years	Linux	9+ years
C++	7+ years	Drivers	9+ years
Shell/Bash	9+ years	DevOps/CI Automation/Tooling	6+ years
Yocto	6+ years	Buildroot	7+ years
Computer Vision, AI Model Training AI Model Inference YOLO, HRnet	1+ years	AOSP Bringup and customizations	2+ years

Relevant recent work experience in the Yocto space:

- Create and maintain some of the BSPs here (mostly at Dunfell):
 - <https://github.com/mxt-creation?q=rzv&type=all&language=&sort=>
- Created a migration of the above BSPs into Scarthgap:
 - <https://github.com/shruggie-ro/mxt-rz-bsp>

Relevant recent work experience in the AOSP space:

- Integrate an 80GB zip file of Android 12 from Amlogic into a proper BSP for a Smart TV (circa 2022)
- Bring-up and customizations Android 14 on a Qualcomm QC56490
 - Bootloader and kernel changes from Android 10
 - frameworks/base (and other AOSP framework changes) to disable stuff for security
 - This is for a set of payment devices (POS, Register, Kiosk)

Relevant recent work experience in the networking space:

- Worked on getting Quectel BG95 working on a Linux systems using AT commands:
 - BG95 is a cheaper version than BG96
 - QMI support is limited/incomplete for BG95, so AT commands must be used
 - Had to write an entire state-machine just to get it to work and get unstuck (via power-reset) when the BG95 doesn't respond `\_(ツ)\_/`
 - All sort of workarounds....
 - Ask me for more details at your own risk (it may be difficult to get me to stop :p)

Relevant OpenSource contributions for Ethernet/Switch

- ADIN1200/1300 Industrial Ethernet PHY driver (~2019):
 - <https://github.com/torvalds/linux/commits/master/drivers/net/phy/adin.c>
- ADIN1100 10Base-T1L Industrial Ethernet PHY driver (~2020):
 - <https://github.com/torvalds/linux/commits/master/drivers/net/phy/adin1100.c>
 - In this case, I wrote the initial driver, but a colleague sent it upstream

Professional Experience

May 2022 - Present

Contractor for Embedded/Linux projects

Role: Contractor

Company: Shruggie

Location: Cluj-Napoca

Clients I've had along this trip:

- Block/Square: Android 14 bring-up and customizations for payment devices (POS, cash register, kiosk)
- MXT Creation - Create BSPs for Renesas RZ/V DRP AI SoC families (RZV2L, RZV2M(A))
 - Yocto, Python, Shell, C, MIPI CSI2 camera sensor drivers
- STIHL: features and performance improvements for the iMOW (lawn-mower robot) using C++
- Ateame: rework of a TS muxer library for a HW video encoder using C++
- Allegro DVT (through Deviqon Labs): emulator for HW IP designs using C++
- BrightGrove (consultancy group): develop platform for a new line of smart TVs (for a customer of BrightGrove) using Android 12
- MathWorks (makers of Matlab/Simulink): kernel support for RF applications using C

**March 2021 -
February 2023**

Deviqon Labs

Role: Senior Software Engineer

Location: Cluj-Napoca

Write software aimed at services for the streaming & television industry

Clients:

- Ateame (acquired Anevia): create and extend software for video/audio/TV streaming and DRM using C++
- Allegro DVT: create emulator for a HW IP designs (for video decoders) using C++

**December 2017 -
February 2021**

Analog Devices Gmbh

Role: Linux Kernel Engineer

Location: Cluj-Napoca

Role includes:

- Work on Linux kernel drivers for Analog Devices parts (ADCs, DACs, accelerometers, IMUs)
- Work and enhancements (to the Linux kernel) to better support Analog Devices parts
- Continuously interact with Linux kernel community by submitting changes to the Linux kernel
- Internal code review for new changes being added to the Analog Devices kernel repository

Technologies used:

- Xilinx Zynq 7 series: ZedBoard, ZC706, ZC702
- Xilinx ZynqMP series: ZCU102, ZU11EG
- Intel/Altera Arria10 SoC
- C for Linux kernel/baremetal drivers
- C++ for Scopy
- DeviceTree for configuring boards for the Linux OS

Links:

<https://github.com/analogdevicesinc/linux/commits?author=commodo>

<https://patchwork.kernel.org/project/linux-iiio/list/?submitter=178229>

2016 - 2022

Co-maintainer of the mstpd package

Role: Co-maintainer

Location: [Github](#)

Co-maintain the mstpd package/code-base (Multiple Spanning Tree Daemon) with other developers

<https://github.com/mstpd/mstpd/commits?author=commodo>

2014 - Present

Open-source contributor to OpenWrt

Role: contributor

Location: [Github](#) / git.openwrt.org

OpenWrt is an embedded Linux distribution for routers.

Technologies used: Shell, Makefile, and C for drivers

Contributor of various patches to core OpenWrt.

<https://github.com/openwrt/openwrt/commits?author=commodo>

Maintainer of a few packages in the packages feed:

<https://github.com/openwrt/packages/commits?author=commodo>

**January 2014 -
December 2017**

Ocedo / Riverbed Technology

Role: Embedded Software Engineer

Location: Cluj-Napoca

[Riverbed](#) acquired Ocedo (which was a small networking startup based in Karlsruhe Germany).

Develop and maintain software modules and drivers for network gateways, switches and access-points.

- Integration of 3G/4G connectivity in gateways
- Integration of Spanning Tree (STP) and Link-Layer Discovery (LLDP) protocols in switches
- Support for some Broadcom switch chips

Technologies used: C for apps and Linux kernel drivers, Shell, Makefile, STP, LLDP, Jenkins

April 2013 - January 2014

Betfair.com

Role: DevOps for internal tooling

Location: Cluj-Napoca

Maintenance & support for internal tools: Gitlab, Perforce, SVN, Jenkins, Atlassian (Confluence, JIRA, Fisheye), Fortify.

Automation of deployment processes using various tools (Jenkins CI, Chef, Puppet).

**November 2012 -
April 2013**

EBS Romania (currently NTT Romania)

Role: Embedded Software Engineer for Automotive
Location: Cluj-Napoca

Technologies used: C for various SW modules

Develop SW & documentation for Continental Automotive for various work on instrumentation clusters.

**August 2011 -
November 2012**

NSpyre Romania (currently Capgemini Romania)

Role: Software Engineer
Location: Cluj-Napoca

Technologies used: Python, numpy, scipy

Write software modules & documentation for controlling highly complex industrial machines, for ASML
(<https://www.asml.com/en>)

March 2007 - August 2011 National Instruments Romania

Role: Software Engineer
Location: Cluj-Napoca

Technologies used: C for RTOS & drivers, ethernet, TCP/IP

- Primarily worked on Pharlap (RTOS)
- Write drivers for ethernet cards for PXI products for the Pharlap OS
- Develop & maintain the TCP/IP stack on the Pharlap
- Add advanced debugging features to Pharlap OS to better debug network issues
- Maintain LabVIEW network primitives across Windows, MacOS, Linux, Pharlap and VxWorks operating systems

Education

2003 - 2008

Technical University of Cluj-Napoca

Computer Science Department (English Section)

1999 - 2003

"Mihai Eminescu" High School - Satu Mare

Mathematics and Informatics (English Section)